

Hazem Hassan

AI Engineer – LLM & Agentic Systems – Production ML

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PROFESSIONAL SUMMARY

Senior AI Engineer with 9+ years building and shipping production AI systems end-to-end — not notebooks. Recent focus on LLM-powered applications: agentic workflows with LangGraph, RAG systems with LlamaIndex, and a full LLM evaluation/observability stack (Ragas, LangSmith). Strong software-engineering foundation — production Python, clean FastAPI/REST API design, robust error handling, Docker/Kubernetes, CI/CD — with a habit of thinking in latency, cost-per-query, and output-quality metrics. Comfortable owning the full lifecycle: architecture, code, deployment, evaluation, and monitoring.

ROLES AND SPECIALTIES

Roles: Senior ML Engineer, AI Technical Lead, LLM/Agentic Systems Engineer.

Specialties: Agentic Workflows, RAG, LLM Evaluation, Automated Labeling, Human-in-the-Loop Workflows, Active Learning, Weak Supervision, Dataset Quality Assurance, ML APIs, MLOps, Cloud/Edge Deployment.

CORE SKILLS

LLM & Agentic Systems: LangGraph (agent workflows / tool-calling), LlamaIndex (RAG, embedding fine-tuning, retrieval orchestration), Multi-RAG, Embeddings, Reranking, Chunking Strategies, FAISS, Conversation Orchestration, NeMo Guardrails, Human-in-the-Loop / Escalation Routing, Prompt Engineering

LLM Evaluation & Observability: Ragas (component-level retriever/generator eval), LangSmith (tracing + continuous evaluation), MT-Bench (LLM-as-judge), AlpacaEval, LM-Eval Harness, Elo Rating, Perplexity, Concurrency/Latency Benchmarking

Software Engineering: Production Python (expert), Clean REST/API Design (FastAPI), Error Handling, Modular Service Design, Multi-threaded Real-time Systems, C/C++ (intermediate), Git, Linux

Deployment & MLOps: Docker, Kubernetes, CI/CD, Model Serving, Inference Optimization (ONNX, TensorRT), Monitoring & Failure Analysis

Cloud & Data: GCP (Vertex AI), AWS (SageMaker), SQL / PostgreSQL, NoSQL basics, Pandas, NumPy, Large-Scale Data Pipelines

ML/DL: PyTorch, TensorFlow, Scikit-learn, Model Training & Fine-Tuning, Hyperparameter Tuning, Custom Loss Design, Evaluation Metrics, Error Analysis.

Leadership: Technical Leadership, System Design, Cross-functional Collaboration, Mentoring, Code Reviews.

WORK EXPERIENCE

Senior ML Engineer

Beyond AI

March 2024—Present, Remote, EMEA

- Built agentic LLM workflows with LangGraph — modeling agents as directed graphs with message passing, tool-calling, and execution orchestration across nodes — plus custom orchestration and RAG services on Python + FastAPI.
- Developed RAG systems with LlamaIndex: embedding fine-tuning, retrieval orchestration, synthetic QA generation, and multi-modal pipelines over text, charts, and research papers, backed by a FAISS vector store.
- Engineered a long-document RAG path (recursive + sequential summarization) with NeMo Guardrails intent routing between retrieval and summarization, tuning chunk size to hold time-to-first-token under 5 seconds.
- Built an LLM evaluation framework integrating Ragas (component-level retriever/generator scoring — context relevancy/recall, faithfulness, answer relevancy) and LangSmith for tracing and continuous evaluation; added MT-Bench (LLM-as-judge), AlpacaEval, LM-Eval, Elo, perplexity, and a concurrency/latency benchmark measuring per-token timing under load.
- Built Multi-RAG pipelines (multi-source retrieval, hybrid search, reranking), improving Recall@10 ~20% and reducing hallucinations ~30%.

Technical Lead

IDefy (Digital Identity/eKYC)

October 2022—February 2024, Giza, Egypt

- Led digital identity onboarding products (SaaS/SDK/web/mobile), delivering 3 production deployments end-to-end.
- Built validation and quality-control workflows for OCR, face matching, liveness, and identity-verification datasets, improving verification success rate by 20% through data quality tuning and systematic failure analysis.
- Developed configurable OCR/document-processing pipelines that adapted to new structured documents using limited samples, annotation rules, and validation checks, reaching 96% recognition accuracy.
- Analyzed production failure cases across OCR, face verification, and liveness detection to identify noisy data, poor image quality, edge cases, and model retraining opportunities.

Senior ML Engineer (Computer Vision)

RDI

January 2021—October 2022, Giza, Egypt

- Led the delivery of the Sotoor Arabic OCR System, achieving 12% WER on Arabic manuscripts.
- Built document OCR pipeline modules (layout analysis, segmentation, font detection, recognition, denoising, correction), reducing OCR error by ~7% on noisy scans.
- Adapted the K2 Toolkit for OCR training/decoding, improving training efficiency by ~25% and gaining ~4% WER, boosting sequence recognition stability.

ML Researcher

RDI

January 2017—January 2021, Giza, Egypt

- Collaborated on a team building NLP systems for the Arabic language across speech and text.
- Developed and deployed production systems, including Speech Recognition, Optical Character Recognition, Text Diacritization, Voice Conversion, and Quran Recitation Verification.

PROJECTS

Agentic Agent Call Center

- Built an agentic AI call-center assistant using LLM tool-calling and Multi-RAG retrieval to automate intent handling, answer generation, and CRM/ticketing actions.
- Designed a reliable conversation orchestration layer with guardrails, escalation-to-human routing, and quality evaluation to ensure safe and consistent customer support automation.

Fashimi APP — Lead Engineer

- Built an AI try-on platform using LLMs for recommendation + prompt generation and diffusion models for avatar generation and photorealistic try-on rendering.
- Developed a recommendation engine leveraging mood/occasion/climate/user preferences to produce personalized outfit prompts.

Valorant Overlay — Developer

- Designed a real-time multi-threaded system extracting gameplay data from Valorant for esports competition overlays.
- Implemented object detection, pattern recognition, and color analysis, achieving a response latency of 50-72 msec.

EDUCATION

Bachelor of Science in Engineering

Cairo University – Electrical Electronics and Communication Engineering · 2016

PUBLICATIONS

Arabic Multi-Genre Broadcast Speech Recognition — Conference Paper, December 2019 — IEEE ASRU 2019

Prognostic AI for Post-Operative Infections — Conference Paper, November 2023 — FCI Annual Conference

QuranScript Verification System — Conference Paper, November 2016 — Sixteenth ESOLE Conference of Language Engineering

HOBBIES

Swimming

Gaming

Padel